

Project Guide

Phase 1: Dataset Selection

Domain	Required Sheets/Tables
Sales Analytics	Orders, Customers, Products, Returns, Salespersons
Supply Chain	Orders, Inventory, Suppliers, Shipments, Warehouses
Healthcare	Patients, Doctors, Treatments, Hospitals, Billing
Banking	Customers, Accounts, Transactions, Loans, Branches
HR Analytics	Employees, Attendance, Salary, Training, Departments
E-commerce	Customers, Orders, Products, Reviews, Payments
Education	Students, Courses, Faculty, Attendance, Results
Crime Analytics	Crimes, Locations, Police Stations, Victims, Cases
Hotel Analytics	Bookings, Customers, Rooms, Staff, Payments
Aviation Analytics	Flights, Airports, Passengers, Crew, Delays

→ Dataset Requirements

Students must find datasets having:

- ✓ Minimum 5 Sheets/Tables
- ✓ Minimum 5,000 Records
- ✓ Common Key Columns

Phase 2: Excel Work

Task 1: Data Cleaning

Students must:

- Remove duplicates
- Handle blanks
- Trim spaces
- Proper case conversion
- Data validation
- Error checking

Task 2: Excel Functions (Make KPI Using Various Functions)

Apply:

Lookup Functions

- VLOOKUP
- XLOOKUP
- INDEX MATCH

Logical Functions

- IF
- IFS
- AND
- OR

Statistical Functions

- SUMIFS
- COUNTIFS
- AVERAGEIFS

Task 3: Pivot Tables (Based On Your Data Sets)

Exa:

1. Monthly Sales Analysis
2. Region Wise Analysis
3. Product Performance
4. Customer Analysis
5. Top 10 Products
6. Bottom 10 Products

Task 4: Excel Dashboard

KPIs:

- Total Sales
- Total Orders
- Total Customers
- Profit Margin %
- Average Order Value
- Growth %

- Return Rate %

Visuals:

- Bar Chart
- Pie Chart
- Line Chart
- Donut Chart
- Slicers

Phase 3: Power BI Work

Data Modeling (According To Your Data)

Students must:

Create Relationships (According To Your Data)

Fact Table



Dimension Tables

Example:

Orders → Customers

Orders → Products

Orders → Date

Orders → Salesperson

Star Schema Design

Fact Table:

- Sales

Dimension Tables:

- Customer
- Product
- Date
- Region
- Employee

Must Create DAX Measures Tasks

Minimum 20+ Dax Needed

Power BI Multiple Dashboard Pages (Based On Your Data Sets)

Page 1

Executive Dashboard

KPIs:

- Revenue
- Profit
- Orders
- Customers

Page 2

Customer Analytics

Visuals:

- Top Customers
- Customer Segmentation
- Revenue by Customer

Page 3

Product Analytics

Visuals:

- Top Products
- Category Analysis
- Profitability Analysis

Page 4

Geographic Analysis

Visuals:

- Map
- Region Sales
- State Analysis

Page 5

Trend Analysis

Visuals:

- Monthly Trend
- Quarterly Trend
- Yearly Trend

Phase 4: Python Analytics

Libraries:

pandas
numpy
matplotlib
seaborn
plotly

Data Cleaning (If Needed)

Tasks:

`drop_duplicates()`

`fillna()`

`replace()`

`merge()`

`groupby()`

Exploratory Data Analysis

Students should answer:

1. Which product sells most?
2. Which customer generates highest revenue?
3. Which month has highest sales?
4. Which region performs best?
5. Which category is least profitable?
6. Which products have highest return rate?

Visualizations

Create:

Basic

- Bar Chart
- Pie Chart
- Line Chart
- Histogram

Advanced

- Heatmap
- Scatter Plot
- Box Plot
- Correlation Matrix

Phase 5: Business Insights

Students must provide:

Findings

Example:

- West region contributes 35% revenue.
- Product A generates highest profit.
- Customer segment Corporate buys most products.

Recommendations

Example:

- Increase inventory for Product A.
- Focus marketing in West region.
- Reduce stock of low-performing products.

Final Project Submission

Students must submit: (Report Mandatory)

1. Dataset

- Excel File
- Multiple Sheets

2. Excel Dashboard

- Formulas
- Pivot Tables
- Interactive Dashboard

3. Power BI Dashboard

- Data Model
- DAX Measures
- 5 Dashboard Pages

4. Python Project

- Jupyter Notebook
- Charts
- Insights

5. Documentation

- Problem Statement
- Objectives
- Methodology
- Findings
- Recommendations

6. Presentation:

10–15 Slides covering:

- Dataset Overview
- Data Cleaning
- Excel Analysis
- Power BI Dashboard
- Python Insights
- Business Recommendations